

Assignment 4

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May 12, 2026

1 Gradient Computation Check

1.1 Quick Sanity Check on Shapes

After loading the data and implementing the forward and backward propagation functions, I first performed a quick sanity check on the shapes of the trainable parameters $[b, c, U, W, V]$. This ensures that the computed gradients are consistent with their theoretical counterparts in terms of dimensionality.

1.2 Comparison with PyTorch Results

To verify the correctness of the gradient computation, I implemented an equivalent version using PyTorch. The gradients obtained from this approach were compared with those from the self-built function using both absolute and relative error. The results show that the discrepancies are negligible, confirming the correctness of the implementation and thus my gradient computations are bug-free.

```
b: max abs err = 2.776e-17, max rel err = 1.879e-16  
c: max abs err = 4.163e-17, max rel err = 2.828e-16  
U: max abs err = 3.816e-17, max rel err = 5.690e-16  
W: max abs err = 8.674e-18, max rel err = 4.476e-16  
V: max abs err = 3.469e-18, max rel err = 2.188e-16
```

Figure 1: Comparison of gradients

2 Longish training run

After implementing both forward and backward propagation, the model was trained using the Adam optimizer. The smoothed loss throughout training is presented in Figure 2. The model was trained for 4 epochs, corresponding to approximately 177,000 update steps, and the final smoothed loss converged to approximately 1.53.



Figure 2: Smooth loss for 4 epoch run

3 Evolution of Generated Text

The evolution of the text synthesized through the first 100,000 update steps of training process is shown as below:

```

=== iter = 0 (before training) ===
uT1uhPNdD•lad•
l6eSC7l10M-a•iYI•to7.2:(t26uz6An}aLcl,Mrf0jc;EB•3pTtüzrJ;CaEL(k,EYTucmB.i/py?
  jLBKa•BfD9_kgy,UAo"2lFd•OTüi3mRlY7n          uo}!L2GFZL}?Je?gggdo•"Qo
    Ub3Il'.onYK,n9wk(awsZ1dbY;vQA9?N7S3?,EJIhPe M)

=== iter = 10000    smooth_loss = 2.0122 ===
hen.  rabey.
"I juppll Dugha?"
"Heffover,"
"Cecred houliex, dalt aid strminhe lohe miss e bert the rice, mo stan a
  ppattewlrwe foy res ll worms and mo canly neds tadathirg eor!"
    "Yowe to kian it Hir

=== iter = 20000    smooth_loss = 1.8414 ===
Beed! She the cad"
"What," thoughtertien, hew ap ant.  Theye's but - tisont. He cevery were
  fichiod porwarteded ask ererest to driges, Wendons slow Harrioms why's
  hanot, be an Sheing that Sne thonge,

=== iter = 30000    smooth_loss = 1.7805 ===
fers dremes on ho cerped to sore pmeed feecheunder of file, instonuing suring
  had.
"Dobled foor all stand in the like po far tos over orty cornen with natingoss
  wore morsle sest out what yendough they y

=== iter = 40000    smooth_loss = 1.6813 ===
hougw Voldsiden's and fultnge affing acrong thing Voldy the one crotling asto
  and Harry fagnie broughond the like thoughter?"
Harry - wher we."
Harry this you m.
. . . and then - as knew it," hawand

=== iter = 50000    smooth_loss = 1.7410 ===

```

```

coriving mwas fol a Mr. ..."
"Oh, bemath alloove nursirs. Weatling as thunctes and szeptabaging thingm you
grow is monegs to tom spobbory saup. "They're steeeces.
"They wants, colda hagrold coll the

=== iter = 60000    smooth_loss = 1.6559 ===
ry and said in the pleasly if his woed askers, whe squy they, then me evertce,
that has seet, Profceed his keeling, Hogsters. "What extum.
"She dened betad Hogwarts. Seaidling about publenort intornat

=== iter = 70000    smooth_loss = 1.6610 ===
all with Parvedosping on with morains, and him just the prams a-minder he
could you been sullinly hink?"
"Nhe's not all jom of Pomporan to con-serteching Harry hear frossearsine-fifs,
a I term but her

=== iter = 80000    smooth_loss = 1.5817 ===
ion. . . . The disture insime the edes empteth sword formort becausing at the
Matily. Kiut - I forct Flertantee fally, shilled indor sperty batthere
one said been the Duoff 'round and right foress.
"S

=== iter = 90000    smooth_loss = 1.6012 ===
Yough-has, into his, small the wanded to do....Wayle trar for a facles had
dien I swarniereapily down ear arone to ail," said howlet all and then,
Wurfe working, and doored leg's, what he solded
somet

=== iter = 100000    smooth_loss = 1.6513 ===
ringing in Dumbledore sidnog though herase.
"Why barget for alon taulory, I please outis, and Glank on a wimp.
"The stolc and There 'm was for immoring of cogreed there? It will be droure he
secile aga

```

We can observe from the generated text that,

- At the beginning of training, the output is largely nonsensical, with both content and formatting appearing highly disorganized.
- Around 40,000 iterations, recognizable keywords such as “Harry” begin to emerge.
- Starting from approximately 5,000 iterations, the sentence structure becomes more stable, with random line breaks largely disappearing.
- By 10,000 iterations, the use of punctuation marks such as quotation marks and question marks becomes mostly correct.

4 Passage generated from optimal model

Part of the smoothed loss during the 4-epoch training process is shown in Table 1.

Table 1: Model Training Progress (Selected Intervals)

Epoch	Iteration	Smooth Loss	Time (s)
1/4	1,000	3.3217	0.6
1/4	2,000	2.7136	1.3
1/4	3,000	2.3774	1.9
1/4	4,000	2.2591	2.6
1/4	5,000	2.1873	3.2
1/4	6,000	2.1672	3.8
1/4	7,000	2.1489	4.5
1/4	8,000	2.0519	5.1
1/4	9,000	2.0180	5.8
1/4	10,000	2.0122	6.4
... Data omitted for brevity ...			
4/4	171,000	1.5347	167.8
4/4	172,000	1.5274	168.5
4/4	173,000	1.5072	169.2
4/4	174,000	1.4896	169.9
4/4	175,000	1.4644	170.6
4/4	176,000	1.4828	171.3
4/4	177,000	1.5032	172.0

The best-performing model with the lowest loss, is obtained at iteration 175,000, where the smoothed loss reaches 1.4644. This model is therefore selected to generate a 1000-character passage.

I foors of skeam slace, Harry's nots coutiered the shours servalls he cale ugall?"

Cedric snoke he was comions, leet his for because face and clear baited them enautcred of strough from look at pook where had broived. I knelings a down. She langerthataill's wand.

"Do - Hermioned again again, reven't to do you could make Berther namond con't abat Cofugh, it cane when face. "Flateree not and the maze some for voice he daredly," when straugh-pair. The spatepoum stop his him goment, the Meddyoun them all brom gove . . . as up and rold him through hapsen about his knee drew people," said Dumbledore, "what no offates. He was still it would Harry was so boty shose "My wand holded," said Dumbledore starced heingr, and he spulods of the secakeffioture - some of the erungers recondstree, "Moody. The peester eare.

"Yeright belace from.

"And sence it which still," Harry back. She had place was eyes neeverse the mown from Harry closenced, in Harry," Fand it benieved. "Tull bo my made as perf